Research Programme on the Mathematization of Karl Marx's *Capital*

Modules 1 & 2: From the Commodity-Form and General Equivalent Dynamics to Stochastic Circulation and the *Salto Mortale* (Volume I, Chapters 1–3)

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Abstract

This manuscript presents Modules 1 and 2 of a multi-stage research programme dedicated to the rigorous mathematization of Karl Marx's *Capital*. Moving beyond static neo-Ricardian linearism, Module 1 formalizes Volume I, Chapter 1 (*"The Commodity"*) using measure spaces over concrete labor, graded monoids for incommensurable use-values, non-symmetric projection operators for the value-form, fiber-bundle quotient topologies for Commodity Fetishism, and evolutionary potential-game dynamics on the hollow row-stochastic manifold proving the endogenous emergence of the General Equivalent ($G_{\mathrm{I}} \to G_{\mathrm{IV}}$). Module 2 advances to Volume I, Chapters 2 and 3 (*"The Process of Exchange"* and *"Money, or the Circulation of Commodities"*), formalizing the bipartite non-use-value axiom, the stochastic realization filter (*salto mortale*), the non-commutative operator algebra of the circuit $C_i - M - C_j$, Marx's inverted Quantity of Money law, and a continuous-time Markov circulation model proving that hoarding and credit-panic feedbacks generate a saddle-node crisis bifurcation that refutes Say's Law within simple commodity circulation.

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Part I
Module 1: Axiomatics of the Commodity-Form, Real Abstraction, and Graph Dynamics of the General Equivalent (Volume I, Chapter 1)

1 Introduction: Beyond Static Linearism

The mathematical formalization of Marxian political economy has historically been dominated by the neo-Ricardian or Sraffian tradition (e.g., Bortkiewicz, Seton, Morishima, Okishio), which models production through non-negative Leontief matrices and Perron-Frobenius spectral theory. While powerful for analyzing intersectoral reproduction and the transformation problem of Volume III, linear input-output algebra bypasses the foundational logical movement of Volume I, Chapter 1 (*"The Commodity"*).

In Chapter 1, Marx does not presuppose a pre-existing scalar metric of price or commensurability. Instead, he investigates how a **homogeneous metric space (Value)** and a **universal canonical sink (Money)** emerge endogenously out of a **heterogeneous space of incommensurable physical objects (Use-Values) and private concrete labors**. Formalizing this genesis requires mathematical structures capable of representing qualitative incommensurability, real abstraction, polar asymmetry, and topological symmetry-breaking.

2 The Dual Space of the Commodity (Chapter 1, Sections 1 & 2)

A commodity $c \in \mathcal{C}$ is a dual entity possessing **Use-Value** (qualitative, incommensurable) and **Value** (quantitative, commensurable). We represent this duality by constructing two distinct spaces linked by a social abstraction functor.

2.1 The Physical Use-Value Space ($\mathcal{U}$)

Let $\mathcal{T} = \{1, 2, \ldots, n\}$ be the index set of distinct concrete commodity types (e.g., linen, coats, iron, corn). Because distinct use-values lack a common physical metric (1 coat + 1 ton of iron admits no physical scalar sum), the space of physical commodities is constructed as a **disjoint union of 1-dimensional positive cones** (a Graded Monoid):

$$\mathcal{U} = \bigsqcup_{i \in \mathcal{T}} \mathcal{U}_i, \quad \text{where } \mathcal{U}_i = \mathbb{R}_{\geq 0} \cdot \mathbf{e}_i \tag{1}$$

Here, $\mathbf{e}_i$ denotes the basis unit of physical measurement for commodity i (yards, tons, bushels). A specific commodity bundle of type i is written as $q_i \mathbf{e}_i \in \mathcal{U}_i$. There is no native inner product, norm, or addition operator on $\mathcal{U}$ that directly combines or compares $\mathbf{e}_i$ and $\mathbf{e}_j$ for $i \neq j$.

2.2 Concrete vs. Abstract Labor and Socially Necessary Labor Time (SNLT)

Let $\mathcal{L}_i$ represent the set of individual producers (or production processes) creating commodity i. Each producer $\omega \in \mathcal{L}_i$ expends a specific **concrete labor** characterized by technical vectors (skill, intensity, tool efficiency) and requires individual physical clock-time $\tau_i(\omega) \in \mathbb{R}_{>0}$ to produce one physical unit $\mathbf{e}_i$.

Let $(\mathcal{L}_i, \Sigma_i, \mu_i)$ be a probability measure space over the producers of sector i, where $\mu_i(\omega)$ represents the share of total social output of commodity i produced by agent ω, satisfying $\int_{\mathcal{L}_i} d\mu_i(\omega) = 1$.

Definition 2.1 (The Real Abstraction Operator $\mathcal{A}$)**.** *Commodity exchange strips away the qualitative specificity of concrete labor, projecting heterogeneous labor processes onto a single one-dimensional scalar manifold:* ***Abstract Human Labor*** *($\mathcal{V} \cong \mathbb{R}_{\geq 0}$). The unit value λ_i of commodity i—its* ***Socially Necessary Labor Time (SNLT)****—is defined via the expectation operator over the sectoral productivity distribution, weighted by skill/intensity multipliers $\sigma_i(\omega) \in \mathbb{R}_{>0}$:*

$$\lambda_i = \mathcal{A}(\mathcal{L}_i) \coloneqq \int_{\mathcal{L}_i} \sigma_i(\omega) \tau_i(\omega) \, d\mu_i(\omega) \tag{2}$$

Remark 2.1 (Comparative Statics of Technological Shock)**.** *If a technological revolution occurs (e.g., power looms replace handlooms), the measure μ_i shifts mass toward producers with lower $\tau_i(\omega)$. This immediately depresses λ_i across the entire existing inventory of commodity i, capturing Marx's insistence that value is a contemporaneous social relation rather than a historical accounting of past effort.*

The **Total Value Functional** $V : \mathcal{U} \to \mathbb{R}_{\geq 0}$ acting on a homogeneous commodity bundle $q_i \mathbf{e}_i \in \mathcal{U}_i$ is therefore:

$$V(q_i \mathbf{e}_i) = q_i \lambda_i \tag{3}$$

3 The Algebra of the Value-Form (Chapter 1, Section 3)

Marx demonstrates that value (λ_i) cannot appear directly as an isolated scalar; a commodity cannot express its value in its own physical body. Value requires a **polar relational mapping** between two distinct commodities.

3.1 The Value Equivalence Relation ($\sim_V$)

Define a binary relation $\sim_V$ on the disjoint space $\mathcal{U}$:

$$q_i \mathbf{e}_i \sim_V q_j \mathbf{e}_j \iff V(q_i \mathbf{e}_i) = V(q_j \mathbf{e}_j) \iff q_i \lambda_i = q_j \lambda_j \tag{4}$$

While $\sim_V$ is mathematically an equivalence relation (reflexive, symmetric, transitive) on the underlying scalar space $\mathcal{V}$, **its phenomenal expression in the commodity world is strictly non-symmetric**. In the value equation "x commodity A is worth y commodity B," A and B occupy polarized structural roles.

3.2 The Non-Symmetric Value Expression Operator ($\mathcal{E}$)

Definition 3.1 (Relative-Equivalent Operator)**.** *For $i \neq j$, define the operator $\mathcal{E} : \mathcal{U}_i \times \{\mathbf{e}_j\} \to \mathcal{U}_j$, where commodity i occupies the **Relative Form** (the active subject seeking value-expression) and commodity j occupies the **Equivalent Form** (the passive physical mirror whose bodily use-value measures i):*

$$\mathcal{E}(q_i \mathbf{e}_i \mid \mathbf{e}_j) = \left(q_i \frac{\lambda_i}{\lambda_j} \right) \mathbf{e}_j \in \mathcal{U}_j \tag{5}$$

Two fundamental formal properties directly encode Marx's dialectical exposition:

(i) **Irreflexivity of Phenomenal Expression:** For $i = j$, $\mathcal{E}(q_i \mathbf{e}_i \mid \mathbf{e}_i) = q_i \mathbf{e}_i$, which reduces to a meaningless tautology (20 yards of linen = 20 yards of linen) and fails to express value. Valid value-expressions are strictly restricted to off-diagonal pairs $(i, j) \in \mathcal{T} \times \mathcal{T}$ with $i \neq j$.

(ii) **Dimensional Projection:** The codomain of $\mathcal{E}(q_i \mathbf{e}_i \mid \mathbf{e}_j)$ is the physical subspace $\mathcal{U}_j$, *not* abstract labor hours $\mathcal{V}$. Abstract value λ_i becomes phenomenally visible only by projecting i onto the concrete physical metric of j.

3.3 Topological Evolution of the Four Forms of Value

We formalize the progression in Chapter 1, Section 3 as a sequence of **directed graphs** $G_k = (\mathcal{T}, E_k)$, where a directed edge $(i \to j) \in E_k$ signifies that commodity i (Relative Form) expresses its value in the physical body of commodity j (Equivalent Form):

Form of Value	Graph Topology G_k	Algebraic Expression	Structural Defect
I. Elementary / Accidental	**Isolated Dyad:** $E_{\text{I}} = \{(i \to j)\}$	$q_i \mathbf{e}_i \xrightarrow{\mathcal{E}} \left(q_i \frac{\lambda_i}{\lambda_j} \right) \mathbf{e}_j$	*Local & Contingent:* Pairwise ratio only; no global transitivity across $\mathcal{T}$.
II. Total / Expanded	**Out-Star Graph:** $E_{\text{II}} = \{(i \to k) \mid k \neq i\}$	$\mathbf{E}_{\text{II}}(q_i \mathbf{e}_i) = \bigoplus_{k \neq i} \left(q_i \frac{\lambda_i}{\lambda_k} \right) \mathbf{e}_k$	*Incomplete & Heterogeneous:* Non-convergent direct sum of orthogonal metrics.
III. General Form	**In-Star Graph (Sink):** $E_{\text{III}} = \{(k \to u) \mid k \neq u\}$	$\forall k \neq u$: $\mathcal{E}(q_k \mathbf{e}_k \mid \mathbf{e}_u) = \left(q_k \frac{\lambda_k}{\lambda_u} \right) \mathbf{e}_u$	*Symmetry Degeneracy:* Any $u \in \mathcal{T}$ can act as sink; competing sinks block unity.
IV. Money-Form	**Symmetry-Broken Sink:** Unique $m \in \mathcal{T}$ (Gold)	$p_k = \frac{\lambda_k}{\lambda_m}$ (Price scalar in $\mathbf{e}_m$)	*Ideal vs. Realized Value:* Bridges Ch. 1 to Ch. 2–3 (*salto mortale*).

Dialectical Dualization (Form II → Form III): In graph-theoretic and categorical terms, Form III is the **transpose (dual graph)** of Form II: $G_{\text{III}} = G_{\text{II}}^{\top}$. By inverting the rays of the Out-Star so that all $k \in \mathcal{T} \setminus \{u\}$ point into a single excluded commodity u, the disjoint space $\mathcal{U}$ acquires a canonical isomorphism to $\mathbb{R}_{\geq 0}$ mediated by the physical basis unit $\mathbf{e}_u$.

4 Commodity Fetishism as a Quotient Topology (Chapter 1, Section 4)

In Section 4, Marx introduces **Commodity Fetishism**: the social relations among private producers $(\mathcal{L}_1, \ldots, \mathcal{L}_n)$ do not appear directly as transparent social relations; instead, they appear exclusively as quantitative exchange-ratios (prices) attached to things.

We formalize this via an **epimorphism (projection) and a hidden fiber bundle**:

1. Let $\Omega = \prod_{i \in \mathcal{T}} \mathcal{L}_i$ be the **Total Social Labor Configuration Space**, encoding the complete network of the social division of labor, concrete production relations, and productivity distributions μ_i.
2. Let $\mathcal{P} = \mathbb{RP}^{n-1}_{>0}$ be the **Projective Price Simplex** of observable exchange ratios $[p_1 : p_2 : \cdots : p_n]$, where $p_i = \lambda_i / \lambda_m$.
3. The market exchange mechanism acts as a projection map $\Pi : \Omega \to \mathcal{P}$.

Because $\dim(\Omega) \gg \dim(\mathcal{P})$, Π is a non-injective surjection with high-dimensional fibers $\Pi^{-1}(\mathbf{p})$. Private producers in a commodity-producing society cannot observe Ω directly; they receive feedback signals exclusively from $\mathcal{P}$ (fluctuations in relative prices) and reallocate their concrete labor accordingly. **Fetishism** is the structural illusion induced by the projection Π, whereby invariants of the fiber space Ω (social relations of production) are misattributed to the base manifold $\mathcal{P}$ (natural properties of physical commodities).

5 Directed Graph Dynamics: Proving the Emergence of the General Equivalent

In Chapter 1, Section 3 (and Chapter 2, "The Process of Exchange"), Marx identifies a fundamental logical and practical contradiction between the **Expanded Form of Value (G_{II})** and the requirements of a coherent commodity-producing society. If every commodity simultaneously attempts to exclude all others as its private out-star, no commodity is excluded, and social commensurability collapses into $O(n^2)$ non-transitive bilateral ratios. Consequently, the "social action of all other commodities" (*gesellschaftliche Tat*) must spontaneously break the symmetry of the commodity space $\mathcal{T}$ and isolate a unique sink $u^* \in \mathcal{T}$—transforming the collection of Out-Stars (G_{II}) into a single **In-Star topology (G_{III})**.

We formulate and prove this transition as a theorem in non-linear evolutionary graph dynamics and potential games.

5.1 The Hollow Row-Stochastic Manifold of Value-Expression Networks

Let $\mathcal{T} = \{1, 2, \ldots, n\}$ $(n \geq 3)$ be the set of distinct commodity types, with positive unit values $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^n_{>0}$.

Definition 5.1 (Value-Expression Adjacency Manifold)**.** *At time $t \geq 0$, each commodity sector $i \in \mathcal{T}$ allocates weights $w_{ij}(t) \in [0, 1]$ across the other $n - 1$ commodities to serve as its **Equivalent Form**. Because the Relative-Equivalent Operator $\mathcal{E}(q_i \mathbf{e}_i \mid \mathbf{e}_j)$ is strictly irreflexive*

($i \neq j$), the weighted directed graph $G(t) = (\mathcal{T}, W(t))$ evolves on the ***hollow row-stochastic manifold****:*

$$\mathcal{W} = \left\{ W \in \mathbb{R}^{n\times n}_{\geq 0} \;\middle|\; w_{ii} = 0 \text{ for all } i \in \mathcal{T},\ \text{and } \sum_{j\neq i} w_{ij} = 1 \text{ for all } i \in \mathcal{T} \right\} \tag{6}$$

Definition 5.2 (Equivalent Prominence / Normalized In-Degree)**.** *For any state $W \in \mathcal{W}$, define the* ***Equivalent Prominence*** *of commodity $j \in \mathcal{T}$ as:*

$$d_j(W) \coloneqq \frac{1}{n-1} \sum_{k\neq j} w_{kj} \in [0,1] \tag{7}$$

The canonical forms of value correspond to distinct geometric loci in $\mathcal{W}$:

- **Form II (Total / Expanded Form):** The maximally symmetric interior centroid W^{II}, where $w^{\mathrm{II}}_{ij} = \frac{1}{n-1}$ for all $i \neq j$, yielding $d_j(W^{\mathrm{II}}) = \frac{1}{n-1}$ for all $j \in \mathcal{T}$.
- **Form III (General Form of Value):** A corner configuration $W^{\mathrm{III}(u^*)}$ indexed by a unique **General Equivalent** $u^* \in \mathcal{T}$, where every ordinary commodity $i \neq u^*$ directs 100% of its value expression into u^*:

$$w^{\mathrm{III}(u^*)}_{iu^*} = 1 \quad (\forall i \neq u^*) \implies d_{u^*}\left(W^{\mathrm{III}(u^*)}\right) = 1, \quad d_k\left(W^{\mathrm{III}(u^*)}\right) = \frac{w_{u^*k}}{n-1} \leq \frac{1}{n-1} \quad (\forall k \neq u^*) \tag{8}$$

5.2 Micro-Foundations and the Global Social Commensurability Potential

When commodity i expresses its value in commodity j ($i \to j$), that expression achieves social commensurability to the extent that other commodities $k \notin \{i, j\}$ also recognize j as their equivalent ($k \to j$). When both $i \to j$ and $k \to j$ hold, commodity j mediates a convergent two-step wedge $i \to j \leftarrow k$ in the same physical subspace $\mathcal{U}_j$:

$$\frac{\mathcal{E}(q_i \mathbf{e}_i \mid \mathbf{e}_j)}{\mathcal{E}(q_k \mathbf{e}_k \mid \mathbf{e}_j)} = \frac{q_i \lambda_i / \lambda_j}{q_k \lambda_k / \lambda_j} = \frac{q_i \lambda_i}{q_k \lambda_k} \tag{9}$$

Thus, the strategic payoff to sector i from allocating weight w_{ij} to equivalent j is given by:

$$u_{ij}(W) = \alpha \sum_{k\notin\{i,j\}} w_{kj} + \eta_j \tag{10}$$

where $\alpha > 0$ measures the **social commensurability coupling** and $\eta_j \geq 0$ represents physical suitability for the equivalent role (divisibility, durability, uniformity).

Lemma 5.1 (Existence of the Social Commensurability Potential)**.** *The n-sector value-expression system on $\mathcal{W} = \prod_{i=1}^{n} \Delta_i$ is an* ***Exact Potential Game*** *with global potential functional $\Phi : \mathcal{W} \to \mathbb{R}$ defined by:*

$$\Phi(W) \coloneqq \frac{\alpha}{2} \sum_{j=1}^{n} \sum_{i\neq j} \sum_{k\notin\{i,j\}} w_{ij} w_{kj} + \sum_{j=1}^{n} \eta_j \sum_{i\neq j} w_{ij} = \frac{\alpha}{2} \sum_{j=1}^{n} \left[\left(\sum_{i\neq j} w_{ij} \right)^2 - \sum_{i\neq j} w_{ij}^2 \right] + \sum_{j=1}^{n} \eta_j \left(\sum_{i\neq j} w_{ij} \right) \tag{11}$$

satisfying $\frac{\partial \Phi(W)}{\partial w_{ij}} = u_{ij}(W)$ for all $i \neq j$.

Proof. Direct partial differentiation of $\Phi(W)$ with respect to w_{ij} ($i \neq j$) isolates the j-th summand and yields $\frac{\partial \Phi(W)}{\partial w_{ij}} = \alpha \sum_{k\notin\{i,j\}} w_{kj} + \eta_j = u_{ij}(W)$. □

5.3 Replicator Dynamics and the Symmetry-Breaking Theorem

Each commodity sector $i \in \mathcal{T}$ updates its equivalent weights $w_{ij}(t)$ via continuous-time multi-population replicator dynamics:

$$\dot{w}_{ij} = w_{ij}\left(u_{ij}(W) - \bar{u}_i(W)\right), \quad \forall i \in \mathcal{T},\ j \neq i, \quad \text{where } \bar{u}_i(W) = \sum_{l \neq i} w_{il} u_{il}(W) \tag{12}$$

Theorem 5.1 (Instability of the Expanded Form G_{II} and Global Convergence to the General Form G_{III})**.** *Assume pure initial symmetry among commodities ($\eta_1 = \eta_2 = \cdots = \eta_n = 0$) and $\alpha > 0$. Under the replicator dynamics on $\mathcal{W}$:*

(1) ***Strict Lyapunov Ascent:*** *$\Phi(W(t))$ is strictly non-decreasing along trajectories:*

$$\frac{d}{dt}\Phi(W(t)) = \sum_{i=1}^{n} \sum_{j \neq i} w_{ij}\left(u_{ij}(W) - \bar{u}_i(W)\right)^2 \geq 0 \tag{13}$$

with equality if and only if W is a stationary point.

(2) ***Linear Instability of the Expanded Form (W^{II}):*** *The symmetric Expanded Form centroid W^{II} ($w_{ij} = \frac{1}{n-1}$) is dynamically unstable. Any perturbation $S_j = \sum_{i \neq j} \epsilon_{ij}$ to the total in-degree of commodity j grows exponentially with strictly positive eigenvalue:*

$$\mu_{star} = \alpha\left(\frac{n^2 - 3n + 1}{(n-1)^2}\right) > 0 \quad (\forall n \geq 3) \tag{14}$$

(3) ***Global Topological Collapse to the In-Star (G_{III}):*** *For almost all initial conditions $W(0) \in \mathrm{int}(\mathcal{W})$, the trajectory $W(t)$ converges as $t \to \infty$ to a corner configuration $W^{III(u^*)}$ where a single commodity $u^* \in \mathcal{T}$ emerges as the unique General Equivalent:*

$$\lim_{t \to \infty} w_{iu^*}(t) = 1 \quad \forall i \in \mathcal{T} \setminus \{u^*\}, \quad \textit{and} \quad \lim_{t \to \infty} d_{u^*}(W(t)) = 1 \tag{15}$$

Proof. **Part (1):** By the chain rule and Lemma 5.1, $\dot{\Phi} = \sum_{i=1}^{n} \sum_{j \neq i} u_{ij} w_{ij}(u_{ij} - \bar{u}_i)$. Subtracting $\sum_{i=1}^{n} \bar{u}_i \sum_{j \neq i} w_{ij}(u_{ij} - \bar{u}_i) = 0$ yields $\dot{\Phi} = \sum_{i=1}^{n} \sum_{j \neq i} w_{ij}(u_{ij} - \bar{u}_i)^2 \geq 0$. By compactness of $\mathcal{W}$ and LaSalle's Invariance Principle, all trajectories converge to the stationary locus $\{\dot{\Phi} = 0\}$.

Part (2): Linearizing around $w_{ij}^{\mathrm{II}} = \frac{1}{n-1}$ with $w_{ij} = \frac{1}{n-1} + \epsilon_{ij}$ ($\epsilon_{ii} = 0$, $\sum_{j \neq i} \epsilon_{ij} = 0$) and defining $S_j = \sum_{k \neq j} \epsilon_{kj}$ ($\sum_{j=1}^{n} S_j = 0$), the perturbed payoff is $\delta u_{ij} = \alpha(S_j - \epsilon_{ij})$. Summing the linearized replicator equation over $i \neq j$ gives:

$$\dot{S}_j = \frac{\alpha}{n-1} \sum_{i \neq j} \left[(S_j - \epsilon_{ij}) - \frac{1}{n-1} \sum_{l \neq i} (S_l - \epsilon_{il})\right] = \frac{\alpha}{n-1}\left[(n-2)S_j - \frac{1}{n-1} S_j\right] = \alpha\left(\frac{n^2 - 3n + 1}{(n-1)^2}\right) S_j \tag{16}$$

Since $n^2 - 3n + 1 \geq 1 > 0$ for all $n \geq 3$, $\mu_{\mathrm{star}} > 0$, proving instability of W^{II}.

Part (3): A rest point W^* is asymptotically stable under replicator dynamics of a potential game if and only if it is a strict local maximizer of $\Phi(W)$. Along any coordinated multi-agent interior perturbation where two agents i_1, i_2 shift weight $\pm\delta$ between two distinct equivalents j_1, j_2, the second variation of Φ is $\delta^2\Phi = 4\alpha\delta^2 > 0$. Thus, no interior or partially mixed multi-sink state can be a local maximum. At pure vertices where each i selects a unique target $f(i) \neq i$ with in-degrees $k_j = |f^{-1}(j)| \in \{0, \ldots, n-1\}$ ($\sum_j k_j = n$), $\Phi(W) = \frac{\alpha}{2} \sum_{j=1}^{n} k_j(k_j - 1)$. By strict convexity of $k(k-1)$, Φ is uniquely maximized when a single sink u^* achieves $k_{u^*} = n-1$, one node $v = f(u^*)$ has $k_v = 1$, and all other $n-2$ nodes have $k_j = 0$, yielding $\max \Phi = \frac{\alpha}{2}(n-1)(n-2)$ at $W^{\mathrm{III}(u^*)}$. □

Corollary 5.1 (Polar Exclusion of the General Equivalent and Transition to the Money-Form G_{IV})**.** *At the General Form attractor $W^{III(u^*)}$:*

(i) ***Exclusion from the Relative Form:*** *For the General Equivalent u^* itself, $u_{u^*v}(W^{III(u^*)}) = \alpha \sum_{k \notin \{u^*, v\}} w_{kv} = 0$ for all $v \neq u^*$, so $\nabla_{\mathbf{w}_{u^*}} \Phi = \mathbf{0}$ (when $\boldsymbol{\eta} = \mathbf{0}$). Thus, u^* is structurally excluded from the active Relative Form; its value can only be read by transposing the In-Star G_{III} back into the Out-Star G_{II}.*

(ii) ***Symmetry Breaking to the Money-Form (G_{IV}):*** *When physical suitability $\boldsymbol{\eta}$ has a unique maximum $\eta_m > \eta_j$ ($\forall j \neq m$, e.g., Gold), the S_n permutation degeneracy is broken, uniquely selecting $u^* = m$ as the global maximizer $\Phi(W^{III(m)}) = \frac{\alpha}{2}(n-1)(n-2) + (n-1)\eta_m$ and establishing the canonical price vector $p_i = \lambda_i / \lambda_m$.*

Part II

Module 2: Stochastic Exchange, the *Salto Mortale*, and the Metamorphosis $C - M - C'$ (Volume I, Chapters 2 & 3)

6 Introduction to Module 2: From Ideal Price to Realized Value

In Chapter 1, the unit value λ_i of a commodity and its money-price $p_i = \lambda_i / \lambda_m$ are derived as **ideal expectations** over the production measure μ_i. However, as Marx transitions to Chapter 2 (*"The Process of Exchange"*) and Chapter 3 (*"Money, or the Circulation of Commodities"*), a crucial structural contradiction emerges:

> *"Money as a measure of value is the necessary form of appearance of the measure of value which is immanent in commodities, namely labour-time... The price of things is an ideal or notional form; they are not yet realized in gold."* (Vol. I, Ch. 3, Sec. 1–2)

A commodity owner brings a physical bundle $q_i \mathbf{e}_i$ to the market bearing **potential (ideal) value** $V_i^{\mathrm{pot}} = q_i \lambda_i$. For private labor to be validated as actual social labor, the commodity must pass through a **stochastic realization filter**—what Marx terms the ***salto mortale* (mortal leap)** of the commodity ($C_i \to M$). Moreover, once exchange is split into two temporally and spatially separated acts—sale ($C_i \to M$) and purchase ($M \to C_j$)—the identity of Say's Law is broken, and **the formal possibility of crisis** arises endogenously within simple circulation.

7 The Mutual Exclusion Principle and Bipartite Exchange Polarity (Chapter 2)

In Chapter 2, Marx establishes that a commodity cannot function as an immediate use-value to its own producer prior to exchange, whereas for its buyer it must function as a concrete use-value.

Definition 7.1 (The Agent-Commodity State Tuple)**.** *Let $\mathcal{A} = \{1, 2, \ldots, N\}$ be the population of private commodity owners. At time t, agent $\alpha \in \mathcal{A}$ is characterized by the state tuple:*

$$\mathcal{S}_\alpha(t) = (\mathbf{q}_\alpha(t),\, m_\alpha(t),\, \mathbf{u}_\alpha) \in \mathcal{U} \times \mathcal{U}_m \times \mathcal{B}(\mathcal{T}_0) \tag{17}$$

where $\mathbf{q}_\alpha(t) = q_{\alpha,i}\mathbf{e}_i \in \mathcal{U}_i$ *is the physical commodity bundle held by* α, $m_\alpha(t) \in \mathbb{R}_{\geq 0} \cdot \mathbf{e}_m$ *is agent* α*'s holding of the Universal Equivalent (Money), and* $\mathbf{u}_\alpha : \mathcal{T}_0 \to \{0,1\}$ *is a binary* ***concrete need indicator*** *over ordinary commodities* $\mathcal{T}_0 = \mathcal{T} \setminus \{m\}$.

Axiom 7.1 (The Non-Use-Value Condition of Commodity Production)**.** *For any commodity producer* $\alpha \in \mathcal{A}_i$ *producing in sector* $i \in \mathcal{T}_0$*:*

$$u_\alpha(i) = 0 \quad \textit{and} \quad \exists j \in \mathcal{T}_0 \setminus \{i\} \textit{ such that } u_\alpha(j) = 1 \tag{18}$$

Remark 7.1. *Axiom 7.1 encodes Marx's proposition that a producer's own commodity has zero immediate use-value for its owner; it exists purely as a bearer of exchange-value (non-use-value to its owner). To realize its use-value,* $q_{\alpha,i}\mathbf{e}_i$ *must be alienated to a distinct agent* β *satisfying* $u_\beta(i) = 1$.

8 The *Salto Mortale* and the Realization Operator (Chapter 3, Section 2a)

In Chapter 1, Socially Necessary Labor Time was defined with respect to **technical production conditions** ($\lambda_i = \int_{\mathcal{L}_i} \sigma_i \tau_i \, d\mu_i$). In Chapter 3, Section 2a (*"The Metamorphosis of Commodities"*), Marx introduces the **second (market-realization) dimension of SNLT**: even if every yard of linen is woven within the technically necessary labor time λ_i, if the "social stomach" cannot absorb the aggregate quantity $Q_i = \sum_{\alpha \in \mathcal{A}_i} q_{\alpha,i}$ at the normal price $p_i = \lambda_i/\lambda_m$, then a portion of society's labor time is rendered superfluous and devalued.

Definition 8.1 (Aggregate Produced Value vs. Stochastic Effective Demand)**.** *For each sector* $i \in \mathcal{T}_0$ *over a market clearing horizon* $[t, t + \Delta t]$*:*

(1) ***Total Produced Ideal Value*** *(*$V_i^{\boldsymbol{prod}}$***):***

$$V_i^{prod} = \lambda_i Q_i = \lambda_i \sum_{\alpha \in \mathcal{A}_i} q_{\alpha,i} \tag{19}$$

(2) ***Stochastic Effective Monetary Demand*** *(*$D_i(\omega_t)$***):*** *Let* $M_i^{circ}(\omega_t) \in \mathbb{R}_{\geq 0}$ *be a random variable on a filtered market probability space* $(\Xi, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ *representing the aggregate quantity of money* $\mathbf{e}_m$ *directed by buyers toward sector* i*. In units of abstract labor time, the* ***socially validated demand for sector*** i *is:*

$$D_i(\omega_t) = \lambda_m M_i^{circ}(\omega_t) \tag{20}$$

Definition 8.2 (The *Salto Mortale* Realization Operator $\mathcal{R}_i$)**.** *The transition from ideal commodity value* C_i *to realized money* M *is governed by a non-linear* ***Realization Operator*** $\mathcal{R}_i : \mathcal{U}_i \times \Xi \to \mathcal{U}_m$*, operating under two dual adjustment regimes:*

(i) ***Quantity-Rationing / Unsold Inventory Regime (Fixed Price*** $p_i = \lambda_i/\lambda_m$***):*** *Each unit of commodity* i *faces a realization survival probability:*

$$\pi_i(\omega_t) = \min\left(1, \ \frac{D_i(\omega_t)}{V_i^{prod}}\right) = \min\left(1, \ \frac{\lambda_m M_i^{circ}(\omega_t)}{\lambda_i Q_i}\right) \in [0,1] \tag{21}$$

The realized value for producer α *holding* $q_{\alpha,i}\mathbf{e}_i$ *is the contraction projection:*

$$V_{\alpha,i}^{real}(\omega_t) = \lambda_m \mathcal{R}_i(q_{\alpha,i}\mathbf{e}_i, \omega_t) = \pi_i(\omega_t) q_{\alpha,i} \lambda_i \leq V_{\alpha,i}^{pot} \tag{22}$$

The residual $(1 - \pi_i(\omega_t))q_{\alpha,i}\lambda_i$ *represents* ***devalued (unvalidated) private labor****.*

(ii) ***Price-Flexibility / Market-Price Divergence Regime:*** *If the entire physical output Q_i clears via an endogenous market price $\tilde{p}_i(\omega_t) = M_i^{circ}(\omega_t)/Q_i$, then the* ***market-validated unit labor value*** *adjusts to:*

$$\tilde{\lambda}_i(\omega_t) = \lambda_m \tilde{p}_i(\omega_t) = \lambda_i \left(\frac{D_i(\omega_t)}{V_i^{prod}} \right) \tag{23}$$

9 Operator Algebra of the Circuit $C_i - M - C_j$

Simple commodity circulation $C_i - M - C_j$ decomposes into two antithetical phases:

- **First Metamorphosis (Sale, $C_i \to M$):** Represented by the stochastic **Alienation Operator** $\hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_m$:

$$\hat{\mathcal{S}}_i(q_i \mathbf{e}_i) = \left(\pi_i(\omega_t) q_i \frac{\lambda_i}{\lambda_m} \right) \mathbf{e}_m \tag{24}$$

- **Second Metamorphosis (Purchase, $M \to C_j$):** Represented by the **Appropriation Operator** $\hat{\mathcal{B}}_j : \mathcal{U}_m \to \mathcal{U}_j$:

$$\hat{\mathcal{B}}_j(m \mathbf{e}_m) = \left(m \frac{\lambda_m}{\lambda_j} \right) \mathbf{e}_j \tag{25}$$

Proposition 9.1 (Topological Asymmetry Between Sale $\hat{\mathcal{S}}_i$ and Purchase $\hat{\mathcal{B}}_j$)**.** *In the Money-Form topology G_{IV} (In-Star with sink m), the operators $\hat{\mathcal{S}}_i$ and $\hat{\mathcal{B}}_j$ exhibit structural polarity:*

(1) ***Sale ($\hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_m$) is a Stochastic Bottleneck:*** *Because $i \in \mathcal{T}_0$ occupies the Relative Form, it possesses no immediate exchangeability ($\pi_i(\omega_t) \leq 1$) and must leap across the directed edge $(i \to m)$ against finite monetary demand M_i^{circ}.*

(2) ***Purchase ($\hat{\mathcal{B}}_j : \mathcal{U}_m \to \mathcal{U}_j$) Possesses Immediate Universal Exchangeability:*** *Because Money m occupies the Universal Equivalent sink, moving from m to any $j \in \mathcal{T}_0$ traverses the transposed Out-Star $G_{IV}^{\top}$, encountering no realization barrier ($\pi_m \equiv 1$ whenever physical inventories $Q_j > 0$ exist).*

Consequently, the composite circuit operator $\hat{\mathcal{M}}_{i \to j} \coloneqq \hat{\mathcal{B}}_j \circ \hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_j$ acts as a non-expansive stochastic contraction in value space:

$$\hat{\mathcal{M}}_{i \to j}(q_i \mathbf{e}_i) = \left(\pi_i(\omega_t) q_i \frac{\lambda_i}{\lambda_j} \right) \mathbf{e}_j \implies \mathbb{E}\Big[V\Big(\hat{\mathcal{M}}_{i \to j}(q_i \mathbf{e}_i) \Big) \Big] = \mathbb{E}[\pi_i] q_i \lambda_i \leq V(q_i \mathbf{e}_i) \tag{26}$$

10 Breaking Say's Law: Continuous-Time Markov Circulation and the Crisis Bifurcation Theorem (Chapter 3, Sections 2b & 3)

In classical political economy (Say, Ricardo, Mill), money is treated as a neutral veil, collapsing $C_i - M - C_j$ into direct barter $C_i \leftrightarrow C_j$ and rendering general overproduction impossible. Marx refutes Say's Law in Chapter 3 by demonstrating that because $C_i \to M$ and $M \to C_j$ are separated by an intermediate holding state M, **any agent who has sold ($C_i \to M$) is under no immediate compulsion to buy ($M \to C_j$)**. Money can arrest its metamorphosis and petrify into a **Hoard** (*Schatzbildung*, Ch. 3, Sec. 3a).

10.1 The Four-State Circulation Process and Marx's Quantity of Money Law

Definition 10.1 (Continuous-Time Circulation Process)**.** *Each unit of social value in the circulation sphere transitions across four macroscopic states* $\Sigma = \{\mathbf{C}, \mathbf{M}_{circ}, \mathbf{M}_{hoard}, \mathbf{K}\}$*:*

(1) $\mathbf{C}$ *(**Commodity Capital on Market**): Waiting to execute the salto mortale* $C \to M$*.*

(2) $\mathbf{M}_{circ}$ *(**Active Circulating Medium**): Money actively executing purchases* $M \to C$*.*

(3) $\mathbf{M}_{hoard}$ *(**Hoarded Money / Liquidity Reserve**): Money withdrawn from active circulation into idle hoards.*

(4) $\mathbf{K}$ *(**Crisis / Defaulted Inventory**): Unsold commodities whose waiting time exceeds their physical or credit-clearing expiration horizon.*

Let $\mathbf{x}(t) = (x_C(t), x_M(t), x_H(t), x_K(t))^\top$ *denote the state distribution, with transition rates:*

- $\gamma_{sale}(x_M) = \beta x_M$*: Rate of commodity realization* $\mathbf{C} \to \mathbf{M}_{circ}$*, proportional to active circulating money* $x_M(t)$ *and turnover velocity factor* $\beta > 0$*.*
- $\mu_{buy} > 0$*: Rate at which active money purchases commodities* $\mathbf{M}_{circ} \to \mathbf{C}$*.*
- $\sigma_{hoard} \geq 0$*: Rate of hoarding withdrawal* $\mathbf{M}_{circ} \to \mathbf{M}_{hoard}$*.*
- $\rho_{dishoard} > 0$*: Rate of dishoarding re-injection* $\mathbf{M}_{hoard} \to \mathbf{M}_{circ}$*.*
- $\delta_{crisis} > 0$*: Expiration / credit-default hazard rate* $\mathbf{C} \to \mathbf{K}$ *when realization stalls.*

Proposition 10.1 (Marx's Law of the Quantity of Circulating Money — Ch. 3, Sec. 2b & 3b)**.** *Let* $P_{sum}(t) = \sum_{i \in \mathcal{T}_0} p_i Q_i(t) = \frac{1}{\lambda_m} \sum_{i \in \mathcal{T}_0} \lambda_i Q_i(t)$ *be the aggregate sum of commodity prices over period* T*, let* $K_{credit}(t)$ *be obligations cleared via mutual bilateral debt cancellation, let* $D_{due}(t)$ *be maturing promissory notes requiring settlement in money as Means of Payment, and let* v *be the average turnover velocity of a unit of money* $\mathbf{e}_m$*. The required active circulating money stock* $M^{active}(t)$ *satisfies:*

$$M^{active}(t) = \frac{P_{sum}(t) - K_{credit}(t) + D_{due}(t)}{v} = \frac{1}{v\lambda_m} \left(\sum_{i \in \mathcal{T}_0} \lambda_i Q_i(t) - \lambda_m K_{credit}(t) + \lambda_m D_{due}(t) \right) \tag{27}$$

Contrary to the neoclassical Quantity Theory ($MV = PY$)*, intrinsic labor values* (λ_i, λ_m) *determine prices* p_i*, which in turn determine* $M^{active}(t)$*; fluctuations in* $M^{active}(t)$ *relative to total bullion* M^{total} *are buffered by the hoard reservoir* $M^{hoard}(t) = M^{total} - M^{active}(t)$*.*

10.2 The Saddle-Node Realization Crisis Theorem

Theorem 10.1 (Breakability of $C - M - C'$ and the Realization Crisis Phase Transition)**.** *Let the total money stock be normalized to* $x_M(t) + x_H(t) = M_0$*, where the liquidity subsystem satisfies:*

$$\dot{x}_M = \rho_{dishoard}(M_0 - x_M) - \sigma_{hoard} x_M \implies x_M^*(h) = \frac{M_0}{1+h}, \quad h := \frac{\sigma_{hoard}}{\rho_{dishoard}} \in [0, \infty) \tag{28}$$

*In steady circulation, the **Global Realization Efficiency** (the fraction of commodities surviving the salto mortale* $C \to M$ *rather than decaying into* $\mathbf{K}$*) is:*

$$\mathcal{E}_{real}(h) = \frac{\gamma_{sale}(x_M^*)}{\gamma_{sale}(x_M^*) + \delta_{crisis}} = \frac{\beta M_0}{\beta M_0 + \delta_{crisis}(1+h)} \tag{29}$$

Suppose hoarding is endogenously coupled to realization failure (flight to liquidity during credit defaults, Ch. 3, Sec. 3b), such that $h(u) = h_0 + \tilde{\xi}u^2$, where $u \coloneqq 1 - \mathcal{E}_{real} \in [0,1]$ is the unrealized crisis share, $h_0 = \sigma_0/\rho_{dishoard} \geq 0$ is baseline hoarding, and $\tilde{\xi} = \xi/\rho_{dishoard} > 0$ is credit-panic sensitivity. Then:

(1) Every equilibrium crisis share $u^* \in (0,1)$ satisfies the transcendental balance equation:

$$L(u) = R(u), \quad \textit{where } L(u) \coloneqq \left(\frac{\beta M_0}{\delta_{crisis}}\right)\frac{u}{1-u} \quad \textit{and} \quad R(u) \coloneqq 1 + h_0 + \tilde{\xi}u^2 \tag{30}$$

*(2) There exists a critical panic threshold $\tilde{\xi}_c > 0$ (determined by the tangency condition $L(u_c) = R(u_c)$ and $L'(u_c) = R'(u_c)$) at which the circulation system undergoes a **saddle-node bifurcation**, generating a stable high-crisis **Monetary Famine / Realization Collapse Attractor** ($u_3^* \to 1$, $x_M^* \to 0$, $x_H^* \to M_0$).*

Proof. Substituting $x_M^* = \frac{M_0}{1+h(u)}$ into the definition of the unrealized crisis share $u = 1 - \mathcal{E}_{\text{real}}(h(u))$ gives:

$$u = \frac{\delta_{\text{crisis}}\left(1 + h_0 + \tilde{\xi}u^2\right)}{\beta M_0 + \delta_{\text{crisis}}\left(1 + h_0 + \tilde{\xi}u^2\right)} \iff \left(\frac{\beta M_0}{\delta_{\text{crisis}}}\right)\frac{u}{1-u} = 1 + h_0 + \tilde{\xi}u^2 \tag{31}$$

establishing $L(u) = R(u)$. On the interval $u \in [0,1)$, $L(0) = 0 < R(0) = 1 + h_0$ and $\lim_{u\to 1^-} L(u) = +\infty > R(1) = 1 + h_0 + \tilde{\xi}$. For small $\tilde{\xi} > 0$, $L(u)$ and $R(u)$ intersect at a unique low-crisis equilibrium $u_1^* \ll 1$ where $L'(u_1^*) > R'(u_1^*)$ (asymptotically stable). However, for sufficiently large capacity ratio $B/\delta_{\text{crisis}} = \beta M_0/\delta_{\text{crisis}}$, varying $\tilde{\xi}$ causes the parabola $R(u)$ to become tangent to $L(u)$ at an interior point $u_c \in (0, 2/3)$ satisfying:

$$\frac{\beta M_0}{\delta_{\text{crisis}}}\frac{u_c}{1-u_c} = 1 + h_0 + \tilde{\xi}_c u_c^2 \quad \text{and} \quad \frac{\beta M_0}{\delta_{\text{crisis}}}\frac{1}{(1-u_c)^2} = 2\tilde{\xi}_c u_c \tag{32}$$

For $\tilde{\xi} > \tilde{\xi}_c$, the system possesses three equilibria $0 < u_1^* < u_2^* < u_3^* < 1$, where u_2^* is an unstable tipping point ($L'(u_2^*) < R'(u_2^*)$) and u_3^* is a locally asymptotically stable **Realization Crisis Attractor** ($L'(u_3^*) > R'(u_3^*)$). If an exogenous demand or credit shock pushes $u(t)$ above u_2^*, self-reinforcing hoarding drains active circulation $x_M \to x_H$, driving the economy to u_3^*. Thus, the temporal separation of sale ($C \to M$) and purchase ($M \to C$) combined with the hoarding function of money is mathematically sufficient to generate endogenous realization crises prior to capitalist production. □

11 Architecture of the Research Programme

Our ongoing formalization of *Capital* proceeds across four interconnected modules:

- **Module 1: Axiomatics of the Commodity & Value-Form Dynamics (Vol. I, Ch. 1).** Complete.
- **Module 2: Stochastic Exchange & The Metamorphosis $C-M-C'$ (Vol. I, Ch. 2–3).** Complete.
- **Module 3: The Valorization Operator & Exploitation $M - C \ldots P \ldots C' - M'$ (Vol. I, Ch. 4–11).** Dynamical systems modeling the contradiction of the General Formula for Capital, the distinction between Labor-Power (a stock commodity with value λ_{lp}) and Living Labor (a flow operator injecting $\Delta V > \lambda_{lp}$), and the extraction of absolute and relative surplus-value.

- **Module 4: Reproduction Schemes, Mean Field Accumulation & Transformation Invariance (Vol. II & Vol. III).** Stochastic optimal control, Fokker-Planck capital distributions, and conservation manifolds linking values to prices of production.